Experiment No.: 06

Title: Compute Statistical Probability Distribution

Batch: B3 **Roll No.:** 16014223083 **Experiment No.:** 6

Aim:

Resources needed: VSCode, Python(NumPy, Pandas, Scikit-learn), Google Colab, Jupyter Notebook

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Pre Requisite: Data Science, Machine Learning

Theory:

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1. Types of Probability Distribution

1. Continuous

2. Discrete

2. Statistical Properties of Probability Distribution

1. **Mean**
2. **Variance**
3. **Skewness**
4. **Standard Deviation**

Algorithm / Procedure

1. Import necessary Python libraries such as NumPy, Pandas, and SciPy.
2. Use previous practical dataset representing a probability distribution.
3. Calculate the mean of the dataset.
4. Compute the variance and standard deviation.
5. Calculate skewness and kurtosis to analyze distribution shape.
6. Visualize the distribution using histograms or probability plots.
7. Interpret the computed statistical properties.

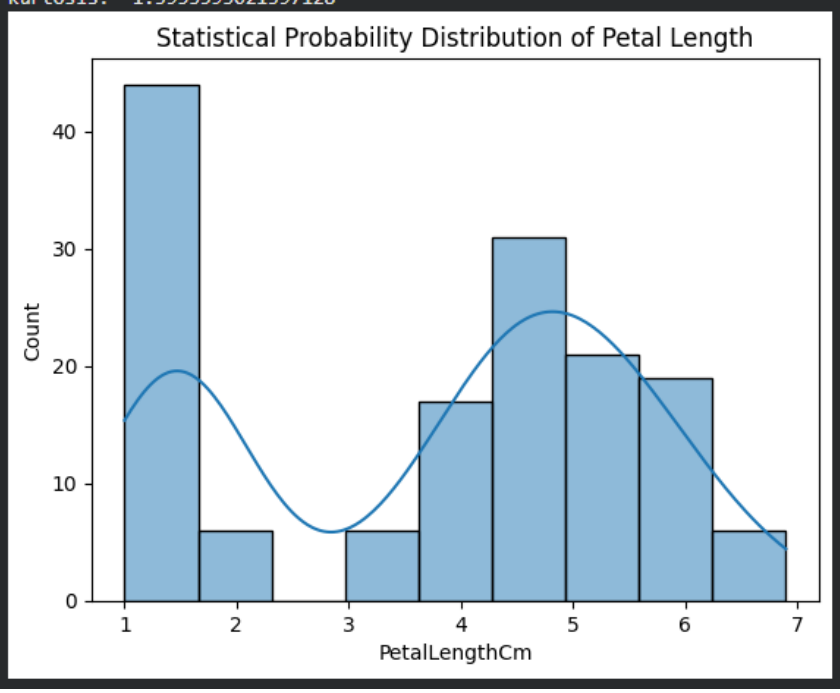
Data Set

Used Datasets from previous Experiment.

Output :

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import skew, kurtosis

# Load dataset (same as previous experiment)
df = pd.read_csv("Iris.csv")

X = df["PetalLengthCm"]

# Statistical Properties
print("Mean:", X.mean())
print("Variance:", X.var())
print("Standard Deviation:", X.std())

# Shape of Distribution
print("Skewness:", skew(X))
print("Kurtosis:", kurtosis(X))

# Visualization
sns.histplot(X, kde=True)
plt.title("Statistical Probability Distribution of Petal Length")
plt.show()
```

```
Mean: 3.758666666666666
Variance: 3.1131794183445156
Standard Deviation: 1.7644204199522617
Skewness: -0.2717119501716388
Kurtosis: -1.3953593021397128
```



Outcomes:

CO4: Apply data analytics to real world applications

Conclusion:

In this experiment, statistical properties of the probability distribution such as mean, variance, standard deviation, skewness and kurtosis were computed using the Iris dataset. The histogram with KDE helped visualize the distribution shape and understand the spread and symmetry of petal length values.

Grade: AA / AB / BB / BC / CC / CD /DD

Signature of faculty in-charge with date

References:

Websites: